

PROJECTION-LATTICE ULTRAFILTERS AT A MEASURABLE CARDINAL

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ABSTRACT. Let κ be a measurable cardinal, $\mathcal{U}$ a normal measure on κ , and $(e_\alpha)_{\alpha < \kappa}$ an orthonormal basis of $\ell^2(\kappa)$. Because every vector of $\ell^2(\kappa)$ has countable support and $\mathcal{U}$ is closed under diagonal intersections, every bounded operator becomes *exactly* diagonal upon compression to $\ell^2(S)$ for a suitable $S \in \mathcal{U}$: Anderson paving with error zero, by a diagonal intersection. We deduce an exact dichotomy — every closed subspace, on a measure-one set of coordinates, either contains the coordinate block or misses it entirely — and conclude that the block filter of a *normal* measure is already a maximal filter in the projection lattice (a *pluf*), κ -complete and nonprincipal — normality, not mere κ -completeness, is what the diagonal-intersection argument consumes, exactly as selectivity, not mere ultrafilterhood, is what maximality consumes at $\aleph_0$. The associated state is a singular, $< \kappa$ -additive pure state. This yields proofs of the existence directions of two theorems of Blecher and Weaver (singular countably additive, respectively $< \kappa$ -additive, pure states on $\mathcal{B}(\ell^2(\kappa))$ exist when κ is Ulam measurable, respectively measurable) that make no use of the Marcus–Spielman–Srivastava solution of the Kadison–Singer problem. In the converse direction, Blecher–Weaver’s excision theory shows that the 1-set of *any* countably additive pure state on $\mathcal{B}(H)$ is a pluf; hence quantum-filter maximality and lattice maximality, which diverge for the singular pure states available below the first Ulam measurable cardinal, coincide above it.

1. INTRODUCTION

This note is the large-cardinal chapter of a study of maximal filters in the projection lattice of a Hilbert space, carried out in the companion papers [5, 6]. There, on a separable space, the maximality of the filter generated by the coordinate subspaces of an ultrafilter turned on delicate matters: it holds exactly for selective ultrafilters [6], the uniqueness of the associated state is the Kadison–Singer problem [9], resolved by Marcus, Spielman and Srivastava [11], and under CH maximal filters exist along which no ellipsoid has round slices [5]. On a Hilbert space of measurable dimension these questions change character, by a mechanism with two ingredients: vectors of $\ell^2(\kappa)$ have countable support, and normal measures absorb diagonal intersections. The rest of the paper follows from the resulting Theorem 2.1.

Our reference point throughout is the paper of Blecher and Weaver [3], which characterizes the cardinals κ such that $\mathcal{B}(\ell^2(\kappa))$ carries singular states with various additivity properties: singular countably additive pure states exist iff κ is Ulam measurable, and singular $< \kappa$ -additive pure states iff κ is measurable. Their existence proofs route purity through a generalization of Kadison–Singer to arbitrary κ ([3], Theorem 5.1), obtained by gluing Marcus–Spielman–Srivastava pavings over all countable subsets of κ by compactness. Section 4 shows that for the existence statements this heavy input can be dispensed with: along a normal measure, purity is a diagonal intersection. The full Blecher–Weaver Theorem 5.1 — unique extension for *arbitrary* pure states on the diagonal, including those living on countable sets of coordinates, where the problem is verbatim the classical one — of course still requires [11]; the boundary between soft and hard runs through the completeness of the ultrafilter, not through the cardinal.

Section 5 runs the bridge in the other direction. On a separable space, the 1-set $\mathcal{F}_\varphi = \{p : \varphi(p) = 1\}$ of a singular pure state is a maximal *quantum filter* in the sense of Farah–Weaver [4] but never a lattice filter: it is not closed under meets, which is precisely why the non-diagonalizable pure states of Akemann–Weaver and Koszmider produce no maximal lattice filters [5]. Blecher–Weaver’s regularity and excision theory ([3], §4) shows that countable additivity repairs this completely: the 1-set of a countably additive pure state is a pluf (Theorem 5.1). Since singular countably additive pure states exist exactly at Ulam measurable dimension, the divergence of quantum-filter maximality from lattice maximality is a phenomenon of small dimension.

Conventions. κ is an uncountable cardinal; a *normal measure* on κ is a κ -complete nonprincipal ultrafilter $\mathcal{U}$ closed under diagonal intersections: if $X_\alpha \in \mathcal{U}$ for $\alpha < \kappa$ then $\Delta_\alpha X_\alpha = \{\beta : \beta \in X_\alpha \text{ for all } \alpha < \kappa\} \in \mathcal{U}$. Every measurable cardinal carries one. We fix an orthonormal basis $(e_\alpha)_{\alpha < \kappa}$ of $\mathcal{H} = \ell^2(\kappa)$; for $S \subseteq \kappa$, $\mathcal{H}_S = \overline{\text{span}}\{e_\alpha : \alpha \in S\}$. A *pluf* is a maximal filter in the lattice $\mathcal{P}(\mathcal{H})$ of closed subspaces; a pluf is *principal* if it consists of the closed subspaces containing a fixed line, and by [5], Lemma 2.2, a pluf containing any finite-dimensional subspace is principal. A filter is maximal iff every closed subspace either belongs to it or meets some member trivially ([5], Lemma 2.1). A state on $\mathcal{B}(H)$ is *singular* if it annihilates the compact operators; $< \kappa$ -additivity of states is as in [3].

2. EXACT PAVING BY DIAGONAL INTERSECTION

Theorem 2.1. *Let $\mathcal{U}$ be a normal measure on κ and $T \in \mathcal{B}(\ell^2(\kappa))$. Then there is $S \in \mathcal{U}$ such that $P_S T P_S$ is diagonal: $\langle T e_\alpha, e_\beta \rangle = 0$ for all distinct $\alpha, \beta \in S$.*

Proof. Each vector of $\ell^2(\kappa)$ has countable support, so $c_\alpha := \text{supp}(T e_\alpha) \cup \text{supp}(T^* e_\alpha)$ is countable and $\kappa \setminus c_\alpha \in \mathcal{U}$ by κ -completeness. Let $S = \Delta_{\alpha < \kappa} (\kappa \setminus c_\alpha) \in \mathcal{U}$. For $\alpha < \beta$ in S we have $\beta \notin c_\alpha$, whence $\langle T e_\alpha, e_\beta \rangle = 0$ and $\langle T e_\beta, e_\alpha \rangle = \langle e_\beta, T^* e_\alpha \rangle = \overline{\langle T^* e_\alpha, e_\beta \rangle} = 0$. $\square$

This is Anderson’s paving [2] with error zero: not small norm on finitely many diagonal pieces, but vanishing off-diagonal on a single measure-one piece. Its spectral consequence:

Corollary 2.2. *Let $T \in \mathcal{B}(\ell^2(\kappa))$, $\epsilon > 0$, and $L = \lim_{\mathcal{U}} \langle T e_\alpha, e_\alpha \rangle$ (the $\mathcal{U}$ -limit exists since T is bounded). Then there is $S \in \mathcal{U}$ with*

$$|\langle T x, x \rangle - L \|x\|^2| \leq \epsilon \|x\|^2 \quad \text{for all } x \in \mathcal{H}_S.$$

For self-adjoint T this says $T_{\mathcal{H}_S}$ is diagonal with all eigenvalues in $[L - \epsilon, L + \epsilon]$.

Proof. Intersect the S of Theorem 2.1 with $\{\alpha : |\langle T e_\alpha, e_\alpha \rangle - L| \leq \epsilon\} \in \mathcal{U}$. For $x = \sum_{\alpha \in S} c_\alpha e_\alpha$ the off-diagonal terms of $\langle T x, x \rangle$ vanish, leaving $\sum_{\alpha} |c_\alpha|^2 \langle T e_\alpha, e_\alpha \rangle$, which lies within $\epsilon \sum_{\alpha} |c_\alpha|^2$ of $L \|x\|^2$. Self-adjointness is nowhere used: the quadratic form sees only the symmetric part of T . $\square$

3. THE SUBSPACE DICHOTOMY AND MAXIMALITY

Theorem 3.1. *Let $\mathcal{U}$ be a normal measure on κ . For every closed subspace $W \subseteq \ell^2(\kappa)$ there is $S \in \mathcal{U}$ with either $\mathcal{H}_S \subseteq W$ or $W \cap \mathcal{H}_S = 0$.*

Proof. Apply Theorem 2.1 to $T = P_{W^\perp}$, obtaining $S_0 \in \mathcal{U}$ with $\langle P_{W^\perp} e_\alpha, e_\beta \rangle = 0$ for distinct $\alpha, \beta \in S_0$. Write $g_\alpha = P_{W^\perp} e_\alpha$; then for distinct $\alpha, \beta \in S_0$,

$$\langle g_\alpha, g_\beta \rangle = \langle P_{W^\perp} e_\alpha, P_{W^\perp} e_\beta \rangle = \langle P_{W^\perp} e_\alpha, e_\beta \rangle = 0 :$$

the family $(g_\alpha)_{\alpha \in S_0}$ is *orthogonal*. Hence for $x = \sum_{\alpha \in S_0} c_\alpha e_\alpha \in \mathcal{H}_{S_0}$,

$$\|P_{W^\perp} x\|^2 = \sum_{\alpha \in S_0} |c_\alpha|^2 \|g_\alpha\|^2,$$

with no cross terms, so $x \in W$ iff $c_\alpha = 0$ whenever $g_\alpha \neq 0$. Writing $S_1 = \{\alpha \in S_0 : g_\alpha = 0\} = \{\alpha \in S_0 : e_\alpha \in W\}$, this says $W \cap \mathcal{H}_{S_0} = \mathcal{H}_{S_1}$ exactly. Now $\mathcal{U}$ decides: if $S_1 \in \mathcal{U}$, take $S = S_1$ and $\mathcal{H}_S \subseteq W$; otherwise $S = S_0 \setminus S_1 \in \mathcal{U}$ and $W \cap \mathcal{H}_S = 0$. $\square$

Remark 3.2. On a measure-one set, incidence and angle coincide: W and the coordinate blocks are in exact general position. This is the geometric content of the theorem, and the reason the pathologies of [5, 6] — all of which stem from subspaces meeting a block trivially while lying at angle zero to it — vanish along $\mathcal{U}$. The contrast with the separable dichotomy of [6], Corollary 2.2 is sharp: there the surviving alternative is *finite codimension in $\mathcal{H}_S$* ; here it is containment of $\mathcal{H}_S$ on the nose, and no auxiliary finite-codimension subspaces need be adjoined to reach maximality.

Corollary 3.3. $\Phi(\mathcal{U}) := \{M \in \mathcal{P}(\ell^2(\kappa)) : \mathcal{H}_S \subseteq M \text{ for some } S \in \mathcal{U}\}$ is a nonprincipal, κ -complete pluf.

Proof. $\Phi(\mathcal{U})$ is a proper filter: $\mathcal{H}_S \cap \mathcal{H}_{S'} = \mathcal{H}_{S \cap S'}$ and $\mathcal{U}$ is a filter of nonempty sets; upward closure is built in; κ -completeness of $\mathcal{U}$ gives closure under intersections of fewer than κ members (the intersection of blocks over sets $S_i \in \mathcal{U}$, $i < \lambda < \kappa$, contains $\mathcal{H}_{\bigcap S_i}$). Maximality: given closed W , Theorem 3.1 either puts $W \in \Phi(\mathcal{U})$ or produces $\mathcal{H}_S \in \Phi(\mathcal{U})$ with $W \cap \mathcal{H}_S = 0$; apply [5], Lemma 2.1. Nonprincipality: were $\Phi(\mathcal{U})$ principal at a line L , then $L \subseteq \mathcal{H}_S$ for all $S \in \mathcal{U}$; but L has countable support c , and $\kappa \setminus c \in \mathcal{U}$ gives $L \cap \mathcal{H}_{\kappa \setminus c} = 0$. $\square$

Corollary 3.4. Define $\varphi_{\mathcal{U}}(T) = \lim_{\mathcal{U}} \langle T e_\alpha, e_\alpha \rangle$. Then $S_{\Phi(\mathcal{U})} = \{\varphi_{\mathcal{U}}\}$ — the state-space face of [5], §3 is a singleton — so $\varphi_{\mathcal{U}}$ is a pure state, and $\Phi(\mathcal{U})$ has the round slice property for every ellipsoid: for every positive invertible T and $\epsilon > 0$ some $M \in \Phi(\mathcal{U})$ has $r(E_T \cap M) < 1 + \epsilon$, with $\lim_{\Phi(\mathcal{U})} T = \varphi_{\mathcal{U}}(T)$.

Proof. Let $\varphi \in S_{\Phi(\mathcal{U})}$ and T self-adjoint. By Corollary 2.2 pick $S \in \mathcal{U}$ with $(L - \epsilon)P_{\mathcal{H}_S} \leq P_{\mathcal{H}_S} T P_{\mathcal{H}_S} \leq (L + \epsilon)P_{\mathcal{H}_S}$ where $L = \varphi_{\mathcal{U}}(T)$; since $\varphi(P_{\mathcal{H}_S}) = 1$, $\varphi(T) = \varphi(P_{\mathcal{H}_S} T P_{\mathcal{H}_S}) \in [L - \epsilon, L + \epsilon]$. As ϵ was arbitrary, $\varphi(T) = L$ for every self-adjoint T : the face is the singleton $\{\varphi_{\mathcal{U}}\}$, whose element, being an extreme point of the state space ([5], Proposition 3.4), is pure. The round-slice statements are the case of positive invertible T , via [5], Lemma 3.1 and Proposition 3.2. $\square$

Remark 3.5 (Scope). The softness is local to the normal measure. An ultrafilter on κ concentrating on a countable subset reproduces the classical Kadison–Singer problem verbatim, and unique extension for arbitrary pure states on the diagonal of $\mathcal{B}(\ell^2(\kappa))$ — Blecher–Weaver’s Theorem 5.1 — still requires [11]. Likewise the general pathology of [5, 6] persists at κ : the pluf space over $\ell^2(\kappa)$ is non-compact, has no prime filters, and admits non-intimate subspaces (graphs of unitaries between complementary blocks), so non-diagonalizable plufs are not excluded. What Theorem 3.1 *does* exclude is any witness family for the reduction of [6], Proposition 5.3 consisting of coordinate subspaces of any single basis.

Remark 3.6 (Normality is the κ -analogue of selectivity). Corollary 3.3 consumes normality, and the parallel with the separable theory is exact. At $\aleph_0$, the block filter of an ultrafilter (with the finite-codimension subspaces adjoined) is maximal if and only if the ultrafilter is *selective* ([6], Theorem 3.4) — and selective ultrafilters are exactly the Rudin–Keisler minimal ones. At κ , the proof of Theorem 2.1 uses the diagonal intersection, i.e. *normality*, not mere

κ -completeness — and normal ultrafilters are exactly the Rudin–Keisler minimal ones among κ -complete ultrafilters on κ , with the Rowbottom partition property playing the role of the Ramsey property. So the same combinatorial regularity (RK-minimality in the appropriate category) powers maximality at both cardinals; the difference is that at κ every measurable cardinal supplies a normal measure outright, while at $\aleph_0$ selectivity is an extra, independence-prone hypothesis. How much of the converse direction of [6], Theorem 3.4 survives at κ is settled in part by the next two results. Every vector of $\ell^2(\kappa)$ — hence every constraint vector of a witness subspace — lives on countably many coordinates, so the witness construction is governed entirely by partitions of κ into *countable* pieces, which are automatically $\mathcal{U}$ -small by κ -completeness. The fibers of a regressive function play no role; the pieces simply *are* the supports.

Definition 3.7. Say a κ -complete nonprincipal ultrafilter $\mathcal{U}$ on κ is a σ -*Q-point* if every partition of κ into countable nonempty pieces admits a partial selector in $\mathcal{U}$ (a set meeting each piece at most once); equivalently, if every countable-to-one function $g : \kappa \rightarrow \kappa$ is injective on a set of $\mathcal{U}$. (For the equivalence: the fibers of a countable-to-one g form such a partition, and injectivity on S is exactly the selector property; in the other direction, send each point to the least element of its piece — a countable-to-one function whose injectivity sets are the selectors.) This is the exact κ -analogue of the Q-point property at $\aleph_0$, with “finite-to-one” promoted to “countable-to-one”.

Lemma 3.8. *Every normal measure is a σ -Q-point. Indeed, for a partition of κ into countable pieces, the set M of piece-minima itself lies in $\mathcal{U}$, and M is a selector.*

Proof. Suppose $M \notin \mathcal{U}$, so $X = \kappa \setminus M \in \mathcal{U}$. The map $g(\alpha) = \min(\text{piece of } \alpha)$ satisfies $g(\alpha) < \alpha$ on X (a point outside M is not the minimum of its piece), so by normality (Fodor) g is constant on a set of $\mathcal{U}$, which is then contained in a single countable piece — impossible for a κ -complete nonprincipal ultrafilter. So $M \in \mathcal{U}$; and $|M \cap \sigma| = 1$ for every piece σ . $\square$

Proposition 3.9. *Let $\mathcal{U}$ be κ -complete and nonprincipal. If $\mathcal{U}$ is not a σ -Q-point, then $\Phi(\mathcal{U})$ is not maximal.*

Proof. Fix a partition $\kappa = \bigsqcup_i \sigma_i$ into countable pieces with no partial selector in $\mathcal{U}$. For each i enumerate $\sigma_i = \{\alpha_{i,n} : n < |\sigma_i|\}$ and set $f_i = \sum_n 2^{-n-1} e_{\alpha_{i,n}} \in \ell^2(\kappa)$: square-summable because the piece is countable, with full support σ_i . Let $W = \{f_i : i\}^\perp$.

Since the pieces are pairwise disjoint and cover κ , every $x \in \mathcal{H}_S$ decomposes as $x = \sum_i x^{(i)}$ with $x^{(i)} \in \ell^2(\sigma_i \cap S)$, and $x \in W$ iff each $\langle x^{(i)}, f_i \rangle = 0$. Three consequences. First, $W \notin \Phi(\mathcal{U})$: membership requires $\mathcal{H}_S \subseteq W$ for some $S \in \mathcal{U}$, i.e. S disjoint from every support — but the supports cover κ . Second, no member of $\Phi(\mathcal{U})$ blocks W : a member contains some $\mathcal{H}_S$, $S \in \mathcal{U}$, and $W \cap \mathcal{H}_S = 0$ holds iff for every i the only vector of $\ell^2(\sigma_i \cap S)$ orthogonal to f_i is zero, iff $|\sigma_i \cap S| \leq 1$ for every i (the coefficients of f_i are nonzero throughout σ_i) — that is, iff S is a partial selector, excluded by hypothesis. So $W \cap \mathcal{H}_S \neq 0$ for every $S \in \mathcal{U}$, and third, W is therefore *addable*: it meets every member of $\Phi(\mathcal{U})$. Addable and unadded: by the maximality criterion ([5], Lemma 2.1), $\Phi(\mathcal{U})$ is not maximal. $\square$

Corollary 3.10. $\Phi(\mathcal{U})$ maximal $\Rightarrow \mathcal{U}$ is a σ -Q-point; and normal $\Rightarrow \sigma$ -Q-point (Lemma 3.8). The characterization question is thus pinched between the two: whether the σ -Q-point property is also sufficient for maximality, and whether it is strictly weaker than isomorphism to a normal measure, are Question 6.3.

Remark 3.11 (Formal verification). Every lattice-level assertion of this paper has been formalized and machine-checked in Lean 4 with Mathlib (toolchain v4.28.0): Theorem 2.1 (exact

paving); Corollary 2.2 in the quadratic-form statement given above, together with the existence of ultrafilter limits of bounded functions; Theorem 3.1 (the exact dichotomy); Corollary 3.3 in full filter language (the block filter is proper, upward closed, closed under intersections — completeness being certified generically in the index type, so that σ - and $<\kappa$ -completeness are instances — nonprincipal, and decides every closed subspace); Lemma 3.8 in the abstract form (Fodor property plus the absence of countable measure-one sets imply the σ -Q-point property, the piece-minima forming a measure-one selector); and Proposition 3.9 in both halves — the combinatorial (σ -Q equivalences and the transversal lemma) and the Hilbert-space (the witness subspace of a countable-piece partition, unadded yet meeting every member of the block filter). The mechanized development also certifies that a block meets the witness trivially exactly when its index set is a partial selector; the hypothesis that the pieces cover κ is necessary there, and a counterexample without it is certified alongside. The artifacts compile with no unproved obligations, and every statement's axiom footprint is exactly **propext**, **Classical.choice**, **Quot.sound**. No large-cardinal machinery is axiomatized: closure under diagonal intersections, countable completeness, and the smallness of countable sets enter only as hypotheses, so the mechanized statements are theorems of ZFC about arbitrary ultrafilters on a (for the combinatorics, well-ordered) index type. Sections 4–5 are likewise machine-checked, with the excision and regularity theory of [3] quarantined as a named package of hypotheses — σ -filtration of the 1-set and the existence of excising projections — and with no appeal, direct or indirect, to [11]: the ultrafilter-limit state is verified to be a state, to annihilate the compact operators, to be countably additive (and additive in the generic-index form that specializes to $<\kappa$ -additivity), and to be the unique element of the face of $\Phi(\mathcal{U})$, whence pure; the 1-set of a state satisfying the excision package is verified to be a pluf, nonprincipal when the state annihilates the finite-rank projections; and the 1-set of the limit state is verified to be exactly $\Phi(\mathcal{U})$. The formalization is carried out over the reals throughout; no argument of these sections required complex scalars. The verification artifacts will be archived with the final version of this paper.

4. EXISTENCE OF SINGULAR ADDITIVE PURE STATES, WITHOUT PAVING

Theorem 4.1. *Let κ_0 be measurable with normal measure $\mathcal{U}$, and let $\kappa \geq \kappa_0$. Regard $\ell^2(\kappa_0) \subseteq \ell^2(\kappa)$ as a coordinate block. Then*

$$\varphi(T) = \lim_{\alpha \rightarrow \mathcal{U}} \langle T e_\alpha, e_\alpha \rangle \quad (T \in \mathcal{B}(\ell^2(\kappa)))$$

is a singular, countably additive pure state on $\mathcal{B}(\ell^2(\kappa))$; for $\kappa = \kappa_0$ it is $<\kappa_0$ -additive. In particular the existence directions of Theorems 2.4(ii) and 2.4(iv) of [3] hold with no appeal to the Kadison–Singer theorem of [11].

Proof. φ is a state (a $\mathcal{U}$ -limit of vector states). Diagonal intersections over $\alpha < \kappa_0$ of the countable supports $c_\alpha \subseteq \kappa$ go through verbatim ($c_\alpha \cap \kappa_0$ is countable, hence κ_0 -small), so Theorems 2.1 and 3.1 hold for blocks indexed by $S \in \mathcal{U} \subseteq \mathcal{P}(\kappa_0)$ inside $\mathcal{P}(\ell^2(\kappa))$, and Corollaries 3.3 and 3.4 give: $\Phi(\mathcal{U})$ is a pluf in $\mathcal{P}(\ell^2(\kappa))$ with $S_{\Phi(\mathcal{U})} = \{\varphi\}$, so φ is pure.

Singularity. A rank-one projection P_v has $\langle P_v e_\alpha, e_\alpha \rangle = |\langle v, e_\alpha \rangle|^2$, nonzero for only countably many α ; κ_0 -completeness gives $\varphi(P_v) = 0$, and singularity follows since singular states are exactly those vanishing on all rank-one projections.

Countable additivity. Let (q_n) be orthogonal projections. At most countably many $\varphi(q_n)$ are nonzero and $\sum \varphi(q_n) \leq 1$; splitting off that part and using finite additivity, it suffices to treat $\varphi(q_n) = 0$ for all n . Given $\epsilon > 0$, the sets $A_n = \{\alpha : \langle q_n e_\alpha, e_\alpha \rangle < \epsilon 2^{-n}\}$ lie in $\mathcal{U}$ (their

complements would give $\varphi(q_n) \geq \epsilon 2^{-n}$, and $\bigcap_n A_n \in \mathcal{U}$ by countable completeness; on it $\langle (\sum q_n)e_\alpha, e_\alpha \rangle = \sum_n \langle q_n e_\alpha, e_\alpha \rangle < 2\epsilon$, so $\varphi(\sum q_n) \leq 2\epsilon$ for all ϵ , i.e. $= 0$.

< κ_0 -additivity when $\kappa = \kappa_0$. By [3], Proposition 2.2(iii) it suffices to check normality of φ on atomic abelian subalgebras generated by fewer than κ_0 orthogonal projections, which is the transfinite monotone convergence Lemma 3.2 of [3] applied to the κ_0 -additive measure $\mathcal{U}$; alternatively, run the argument of the previous paragraph with $(q_\beta)_{\beta < \lambda}$, $\lambda < \kappa_0$, replacing the sets A_n by $A_\beta^\epsilon = \{\alpha : \langle q_\beta e_\alpha, e_\alpha \rangle < \epsilon_\beta\}$ for any summable (ϵ_β) supported on the countably many relevant β — for each fixed α , $\langle q_\beta e_\alpha, e_\alpha \rangle \neq 0$ for only countably many β , and κ_0 -completeness applies. $\square$

Remark 4.2. For Ulam measurable κ (i.e. $\kappa \geq$ some measurable κ_0) the theorem produces the singular countably additive pure state on $\mathcal{B}(\ell^2(\kappa))$ required by [3], Theorem 2.4(ii); the reverse directions of the Blecher–Weaver characterizations, being nonexistence statements, are untouched and remain as proved there. What the diagonal intersection replaces is exactly the compactness gluing of uncountably many Marcus–Spielman–Srivastava pavings in their Theorem 5.1 — for those pure states on the diagonal that come from κ -complete ultrafilters. The hard/soft boundary in Kadison–Singer at large κ thus runs through completeness, not cardinality.

5. COUNTABLY ADDITIVE PURE STATES GIVE PLUFS

We now use Blecher–Weaver’s own machinery to close the circle. Recall ([3], §4): a state φ on a von Neumann algebra is *regular* if countable joins of null projections are null; every countably additive state is regular, and regular *pure* states are countably additive ([3], Theorem 4.1, after [8]); for regular φ the 1-set $\mathcal{F}_\varphi = \{p : \varphi(p) = 1\}$ is a σ -filter ([3], Lemma 4.3); a projection $p \in \mathcal{F}_\varphi$ *excises* φ for x if $pxp = \varphi(x)p$, and for countably additive pure φ every x admits an excising projection in $\mathcal{F}_\varphi$ ([3], Lemma 4.5); finally, a pure state is determined by its 1-set ([3], Lemma 4.2).

Theorem 5.1. *Let φ be a countably additive pure state on $\mathcal{B}(H)$, H of arbitrary dimension. Then $\mathcal{F}_\varphi$ is a pluf (identifying projections with their ranges), and $S_{\mathcal{F}_\varphi} = \{\varphi\}$. If φ is singular, $\mathcal{F}_\varphi$ is nonprincipal.*

Proof. $\mathcal{F}_\varphi$ is a filter: closed under (countable) meets by regularity ([3], Lemma 4.3), upward closed, and proper since $\varphi(0) = 0$. Maximality: let M be a closed subspace with $P_M \notin \mathcal{F}_\varphi$, i.e. $c := \varphi(P_M) < 1$. By excision ([3], Lemma 4.5) choose $p \in \mathcal{F}_\varphi$ with $pP_M p = cp$. If v were a unit vector of $\text{ran } p \cap M$, then $1 = \langle P_M v, v \rangle = \langle pP_M p v, v \rangle = c < 1$, absurd; so $\text{ran } p \cap M = 0$, and [5], Lemma 2.1 gives maximality. That $S_{\mathcal{F}_\varphi} = \{\varphi\}$ is [3], Lemma 4.2. If φ is singular it vanishes on every rank-one projection, so $\mathcal{F}_\varphi$ contains no line; a principal pluf contains a line ([5], Lemma 2.2), so $\mathcal{F}_\varphi$ is nonprincipal. $\square$

Corollary 5.2. *For countably additive pure states on $\mathcal{B}(H)$, the maximal quantum filter $\mathcal{F}_\varphi$ of Farah–Weaver [4] is a maximal lattice filter: quantum-filter maximality and lattice maximality coincide. By contrast, for singular pure states on a separable H — which are never countably additive — the 1-set is not even a filter ([4]; [3], remark before Lemma 4.3), which is why the non-diagonalizable pure states of [1, 10] yield no plufts [5]. Since singular countably additive pure states exist exactly when $\dim H$ is Ulam measurable ([3], Theorem 2.4(ii)), the divergence of the two maximality notions for singular states is a phenomenon of dimension below the first Ulam measurable cardinal.*

Corollary 5.3. *In the setting of Corollary 3.4, $\mathcal{F}_{\varphi_{\mathcal{U}}} = \Phi(\mathcal{U})$: the pluf is exactly the 1-set of its state.*

Proof. $\Phi(\mathcal{U}) \subseteq \mathcal{F}_{\varphi_{\mathcal{U}}}$ since $\varphi_{\mathcal{U}}(P_M) = 1$ for $M \in \Phi(\mathcal{U})$; both are plufs (Corollary 3.3, Theorem 5.1); a maximal filter admits no proper filter extension. $\square$

6. QUESTIONS

Question 6.1. Anderson’s conjecture at κ : is every countably additive pure state on $\mathcal{B}(\ell^2(\kappa))$ of the form $\varphi_{\mathcal{U}} \circ (\text{basis change})$ for some κ -complete ultrafilter and orthonormal basis — equivalently (via Theorem 5.1 and [6], Theorem 5.1), is every pluf of the form $\mathcal{F}_{\varphi}$, φ countably additive, diagonalizable? Blecher–Weaver note the conjecture is expected to be consistent; a counterexample would produce a non-diagonalizable pluf with singleton state face, answering the quadrant question of [5] at κ . The refutations known at $\aleph_0$ [1, 10] concern singular non-regular states and do not transfer.

Question 6.2. By [3], Theorem 4.6, a $< \kappa$ -additive pure state is normal on every von Neumann subalgebra generated by fewer than κ elements; in particular any separably-acting chain such as $\{L^2[0, t]\}$ is annihilated and invisible to $\mathcal{F}_{\varphi}$. The κ -analogue of the continuous-masa question of [6] is therefore: working in $L^2(\{0, 1\}^{\kappa})$ with the product measure — a κ -generated diffuse masa, outside the reach of their Theorem 4.6 — does the family of spectral subspaces of the masa defeat every basis in the sense of [6], Proposition 5.3? Theorem 3.1 shows any witness family must be transverse to every atomic masa; a diffuse masa is the extreme case.

Question 6.3. The κ -analogue of [6], Theorem 3.4, refined by Proposition 3.9: is $\Phi(\mathcal{U})$ maximal if and only if $\mathcal{U}$ is a σ -Q-point? Necessity is Corollary 3.10; sufficiency is open, and would require an exact dichotomy (Theorem 3.1), or a substitute for it, proved from the σ -Q-point property alone, without diagonal intersections.

Maximality does not, however, characterize normality up to isomorphism. In [7] the Fubini product $\mathcal{D} \otimes \mathcal{D}$ of a normal measure with itself is shown to satisfy exact paving outright — Rowbottom homogeneity for finitely many colorings of triples and quadruples replaces the diagonal intersection — so that $\Phi(\mathcal{D} \otimes \mathcal{D})$ is a κ -complete nonprincipal pluf with singleton face, while $\mathcal{D} \otimes \mathcal{D}$ is isomorphic to no normal measure. The sharper questions are posed there: whether maximality, exact paving and the σ -Q-point property coincide.

Question 6.4. The generalized Calkin algebra $\mathcal{B}(\ell^2(\kappa))/\mathcal{K}$ (or the quotient by operators of support $< \kappa$) carries its own projection lattice. Do the maximal filters there lift, in the sense of Farah–Wofsey’s Question 6.39 [4], and does the exact dichotomy of Theorem 3.1 descend to the quotient along normal measures?

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